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Impact of Microaeration and the Redox Mediator Anthraquinone-2,6-Disulfonate on Azo Dye Reduction and By-Products Degradation

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This work assessed the impact of microaeration and the redox mediator anthraquinone-2,6-disulfonate (AQDS) on both azo dye reduction and by-products degradation (aromatic amines) in a mesophilic reactor. The experiment was carried out in an up-flow anaerobic sludge blanket (UASB) reactor with 3.2 L of working volume and hydraulic retention time (HRT) of 24 h. The synthetic effluent contained 50 mg L^{-1} of the dye Reactive Red 2 (RR2). The reactor was initially operated under anaerobic conditions and, subsequently, was microaerated with atmospheric air (1 mL min^{-1}) and then supplemented with the redox mediator (AQDS) at a concentration of 25 μM. The microaeration effect on dye reduction and by-products formed was not evident. However, AQDS considerably enhanced RR2 reduction, and its regeneration was not affected by the presence of oxygen. It seems that the cleavage of the double bond between nitrogen groups is a much easier way to start the attack of the azo dye, and it is independent of the presence of oxygen. Thus, the introduction of hydroxyl groups into the aromatic ring was not playing a role in the color removal process. In terms of ecological parameters, the ecological niches were maintained, and the distribution of species increased, despite the majority being generalists, i.e., not specialized in dye degradation. Furthermore, anaerobic species are dominant over facultative species, which explains why there was no population diversity recovery after the microaeration started.

1. Introduction

The textile sector is one of the most representative segments in industry due to its importance to the economy of developed countries and because it represents a "flagship" in several developing countries. In 2015, it was responsible for US$ 291 billion of world exports, in which China, the European Union, and India remained the top three exporters of textiles. Altogether, they accounted for 66.4% of world exports.[1] The Brazilian demand for textile products is still high, generating wastewater with high amounts of dyes, organic matter, chemicals, etc., which poses a risk when discharged without treatment into surface water. The difficulty to treat efficiently colored wastewater in treatment plants is a constant in textile industry, as small amounts of dyes and pigments produce intense coloring.[2] It is estimated that 10^9 kg of dyes are produced every year in the world, among which 70% are azo dyes (−N═N−).[3,4] These dyes are molecules soluble in water, which can contain one or more ionic groups, and the most common used are sulfonic acid and amine groups. These are considered recalcitrant and non-biodegradable.[5,6] Large amounts of dyes are lost during industry process and goes to the environment as wastewater. Effluents contaminated with dyes are considered toxic to aquatic environment and symbiosis process, and interfere negatively in photosynthesis and primary production, because transparency is reduced.[5]

Some works show that the application of quinone-based compounds, such as anthraquinone-2-sulfonate (AQS) and anthraquinone-2,6-disulfonate (AQDS), vitamins, such as riboflavin, or even yeast extract can accelerate electron transfer and increase color removal.[7,8] Biological azo dye reduction usually requires anaerobic operation because the redox condition achieved is more effective for color removal, whereas removal or biodegradation of the by-products (aromatic amines) occur commonly in aerobic environment, in which the products are used as sources of carbon, nitrogen, and energy.[9]

The addition of microaeration, i.e., introduction of low oxygen concentrations ($< 1\,L\,O_2\,L^{-1}$ feed),[10] can facilitate the initial degradation of some aromatic compounds, including BTEX (benzene, toluene ethylbenzene, and xylenes), hormones, and pharmaceuticals, because, under microaerobic conditions, some microorganisms can use oxygen to introduce hydroxyl groups into the aromatic ring, as in the classical aerobic pathway, subsequently facilitating cleavage by anaerobic pathway.[11] By

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controlling the oxygen levels, the redox potential could be maintained at the proper levels for dye reduction as well, therefore, both processes could take place in the same reactor.

Additionally, there is a lack of studies investigating the effect of redox mediators on dye reduction under microaerobic conditions, thus accelerating the formation and removal of by-products under such conditions. Another important point to investigate is if the microaerobic environment will interfere the regeneration capacity of the redox mediator, thus affecting negatively the color removal efficiency.

Hence, this work assessed the impact of microaeration and the redox mediator AQDS on both azo dye reduction and by-products degradation (aromatic amines) in a mesophilic up-flow anaerobic sludge blanket (UASB) reactor.

2. Experimental Section

2.1. Reactor

The UASB reactor was made of acrylic plastic and polyvinyl chloride (PVC) connections for sewage, with a working volume and hydraulic retention time of 3.2 L and 24 h, respectively. The inoculum was an anaerobic sludge (≈30 g of volatile suspended solids (VSS) per liter), from a mesophilic UASB reactor treating brewery wastewater. The reactor was completely covered with aluminum foil to avoid phototrophic microorganisms proliferation.

The influent was stored at 4 °C, and the reactor was operated at room temperature of ≈27 °C. The biogas produced was collected and subsequently analyzed by GC (Shimadzu, GC 17A) according to Firmino et al.[12]

At some stages, a microaeration system was operated to create an influx of oxygen by pumping atmospheric air into the reactor through the feeding line at a flow rate of 1 mL min^{-1} (28 °C, 1 atm) (Resun Air Pump AC 1000), which corresponds to an influent flow rate to air flow rate ratio (Q_{inf}/Q_{air}) of 2.4.

2.2. Synthetic Textile Wastewater

The synthetic wastewater was composed of water, an azo dye, a carbon source (electron donor), basal medium (nutrients), a buffer, and a redox mediator (when tested). The dye used in this research was Reactive Red 2 (RR2) (40% purity, Sigma–Aldrich, USA) at a concentration of 50 mg L^{-1}.

The electron donor (≈2 g of chemical oxygen demand (COD) per liter) was ethanol (99.8% purity, Dinâmica, Brazil), and the basal medium composition was prepared according to Firmino et al.[13] The wastewater was buffered with sodium bicarbonate ($NaHCO_3$), 1 g $NaHCO_3$/L synthetic influent to maintain pH ≈ 7. At some stages, AQDS (Sigma–Aldrich) was used as redox mediator at a concentration of 25 μM. The reagents were used as purchased, without further purification.

2.3. Experimental Procedure

This study was carried out in five stages (**Table 1**), including the start-up period (stage I), in which ethanol was the only source of

Table 1. Operational parameters and performance of the reactor over the experimental stages.

Stage	I	II	III	IV	V
Operational parameter					
End of the stage (days)	84	211	260	361	447
COD_{Total} (g L^{-1})	2	2	2	2	2
HRT (h)	24	24	24	24	24
Dye concentration, RR2 (mg L^{-1})	–	50	50	50	50
Redox mediator concentration, AQDS (μM)	–	–	–	25	25
Microaeration (mL min^{-1})	–	–	1.0	1.0	–
Q_{inf}/Q_{air}	–	–	2.4	2.4	–
System performance					
COD influent (mg L^{-1})	2328 (128)	2248 (242)	1529 (214)	2100 (236)	1572 (208)
COD effluent (mg L^{-1})	159 (37)	251 (48)	228 (21)	195 (63)	101 (30)
COD removal (%)	93 (1)	89 (3)	86 (3)	90 (5)	93 (3)
pH	7.16 (0.2)	7.25 (0.4)	7.62 (0.3)	7.85 (0.3)	7.59 (0.2)
T (°C)	27.79				
TA (mg L^{-1})	827 (139)	854	786(23)	820	676 (32)
VFA (mg L^{-1})	99 (62)				
VFA/TA	0.12 (0.06)	0.20	0.06 (0.02)	0.23	0.16 (0.07)
Color influent (mg L^{-1})	–	54 (9)	46 (2)	48 (3)	45 (4)
Color effluent (mg L^{-1})	–	27 (6)	22 (2)	8 (1)	8 (1)
Color removal (%)	–	51 (9)	51 (5)	83 (3)	82 (3)

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carbon and energy. During the experiment, there was no change in the hydraulic retention time (HRT) of 24 h.

Subsequently, after the reactor has reached a steady state, RR2 was introduced at a concentration of 50 mg L^{-1} (stage II). At the next stage (III), the microaeration started at a flow rate of 1 mL min^{-1} of atmospheric air (Q_{inf}/Q_{air} ratio of 2.4).

At the next stage, the redox mediator AQDS was added at a concentration of 25 μM (stage IV). Subsequently, the microaeration was turned off at stage V, but the redox mediator was maintained. The transition between stages was done after checking dye removal stability (variation up to 10%) in the last three analysis points (1 week of operation).[12]

After the continuous-flow experiments, a batch assay was conducted for 10 h, using the same reactor, in order to obtain kinetic data of dye reduction and aromatic amine removal. Thus, a peristaltic pump (Watson Marlow, model 323) was used to recirculate the effluent, 90 rpm, 35 L day^{-1}, in order to have a minimum time outside the reactor.

Along the batch experiment, the feeding pump was turned off, and the pump used was the Watson Marlow. After sludge decantation, the supernatant was discarded, and the reactor was fed with synthetic wastewater. Two batch experiments were conducted, under anaerobic and microaerobic conditions, respectively.

2.4. Analyses

The color analysis was performed three times a week, for the continuous-flow experiment, and every 30 min, for the batch experiment, by spectrophotometry (Hach, DR − 6000). Influent samples were previously diluted (1:5) in water, followed by dilution (1:2) in a phosphate buffer, pH 7 (10.86 g L^{-1} $NaH_2PO_4 \cdot 2H_2O$ and 5.98 g L^{-1} $Na_2HPO_4 \cdot 2H_2O$) and then centrifuged for 2 min at 13 000 rpm (Eppendorf, Mini Spin). The effluent samples were filtered and diluted (1:2) in the phosphate buffer. The absorbance of the dye was read at the wavelength of 513 nm (λ_{max}). Furthermore, a scanning of the samples between the wavelengths 190 and 700 nm (Hach, DR − 6000) was performed.

COD and total alkalinity (TA) were determined according to Standard Methods for the Examination of Water and Wastewater,[14] volatile fatty acids (VFA) were determined using the Kapp titrimetric method and pH was determined by a multiparameter probe (Hanna, H9828).

At the end of each stage, sludge samples were collected in order to analyze changes in microbial population dynamics. Molecular biological analyses were performed as detailed elsewhere,[12] and bacterial, and archaeal community structure were evaluated by parameters, such as evenness/functional organization (F_o), ranged weighted richness (R_r), and Shannon diversity index (H).[15]

2.5. Statistical Methods

The computer program SigmaStat 4.0 was used for the statistical data analysis, in which the Mann–Whitney Rank Sum test, a non-parametric procedure which does not require a specific data distribution, was used to compare the performance of the different stages. The results of the tests were evaluated according to the p-value. If $p \leq 0.05$, the null hypothesis is rejected, i.e., the data groups are considered statistically different.[12]

3. Results and Discussion

3.1. Color Removal under Anaerobic and Microaearobic Conditions

The efficiency of co-substrate removal presented instability at the initial stages due to the adaptation of inoculum microbiota. After 60 days of operation, the efficiency and stability improved probably due to microbial adaption. At this stage, the COD removal efficiency was 93% (Table 1). pH values were between 6.6 and 7.4, demonstrating good operational performance of the anaerobic reactor.[16]

After the start-up (stage I), the dye RR2 was introduced at a concentration of 50 mg L^{-1} (stage II), as shown in Table 1. The average color removal efficiency was ≈51% (Table 1). Initially, the removal efficiency was approximately 60%, then decreased to 40% and, subsequently, increased to 55% (**Figure 1**). The highest efficiency at the initial stage occurred probably due to dye adsorption onto the sludge blanket, which seemed to reach its saturation point around the day 22 of operation. It is worth mentioning that the color removal efficiency obtained was considered low, even though the HRT used was high (24 h). The results are in accordance with Dos Santos et al.,[4] who obtained an efficiency of 56% using an expanded granular sludge bed (EGSB) reactor and RR2 concentration of 1.25 g L^{-1}, although the HRT used was only 10 h.

On the contrary, Dos Santos et al.[17] report an efficiency of 92% and 87% for 0.02 and 0.04 g L^{-1} of RR2, respectively, at a HRT of 10 h. One possibility for the higher efficiency is related to the low dye concentration used in the experiment, which, together with a high biomass concentration, promoted a zero order kinetics and, hence, a high efficiency.[17]

During stage III, when microaeration was applied, the system performance was similar to that at stage II ($p = 0.975$), whose

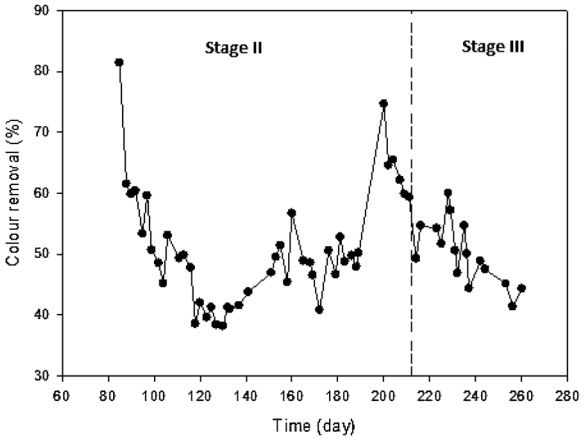


Figure 1. Color removal efficiency during anaerobic (stage II) and microaerobic (stage III) conditions.

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color removal was 51% (Figure 1). Thus, under the conditions tested in the experiment, it was not possible to observe any effect of microaeration on color removal.

Addition of low oxygen concentration can facilitate the initial degradation of some compounds.[11] However, it seems that the cleavage of the double bond between nitrogen groups is a much easier way to start the attack of the azo dye, and it is independent of the presence of oxygen. Thus, the azo bond cleavage was not affected by the presence of oxygen, and the introduction of hydroxyl groups into the aromatic ring, as in the classical aerobic pathway, subsequently facilitating cleavage by anaerobic pathway,[11] was not playing a role in the process. Therefore, microaeration showed to be an inefficient strategy to speed up azo dye reduction.

Microaeration would influence aromatic amines mineralization, helping to break the aromatic ring. However, no remarkable removal was found in the present investigation. Partial or complete azo dye by-product (aromatic amine) removal by aerobic processes is reported in the literature.[9] Therefore, the amount of oxygen must be much higher to increase the oxidation reduction potential (ORP) of the system, and the aromatic amines removal would occur by both cell anabolism and catabolism.

Although the results were unexpected for color and aromatic amines removal, microaeration would be very important to remove other compounds present in textile wastewater, such as adsorbable organic halides (AOX),[18–20] which are recalcitrant under anaerobic conditions.

The absorption spectrum after RR2 influent and effluent scanning during stages II and III is shown in **Figure 2**. According to Feng et al.,[21] who worked with reduction and oxidation of the same azo dye, the absorbance peak at 310 nm are related to naphthalene ring, whereas the peak at 230 nm are related to benzene ring. Both structures exist in the dye, and the divergence between the peaks can be related to groups nearby. Pinheiro et al.[22] mentioned that absorption peaks at 196, 230, and 280 nm are possibly related to aniline, one of the aromatic amines which can be produced from RR2 reduction. Aniline was indeed detected in an EGSB reactor under thermophilic conditions, revealing that this compound could not be further degraded under anaerobic conditions.[21]

Feng et al.[21] showed that, after reductive process, a significant peak at 260 nm appeared, being attributed to the multi-substituted benzene (e.g., benzene dicarboxylic acid) produced from degradation of the naphthalene ring. Nevertheless, in the present experiment, this peak appeared before dye reduction. Feng et al.[21] also showed that, when the azo bond is not destroyed, but hydrogenated (Ar—NH—NH—Ar'), an absorption peak at 245 nm appeared. The absence of this peak reinforces the hypothesis of azo bond cleavage.

A decrease in the visible region around the absorption peak (arrow) at 513 nm was found, showing a reduction in dye concentration (Figure 2). Moreover, RR2 has three absorption peaks at 230 (peak 1), 260 (peak 2), and around 310 nm (peak 3) (Figure 2), which are consistent with the results found by Feng et al.[21] In addition, around 310 nm (peak 3), there was a decrease in the absorption (Figure 2), indicating that, probably, a by-product with naphthalene ring in the molecular structure was removed.

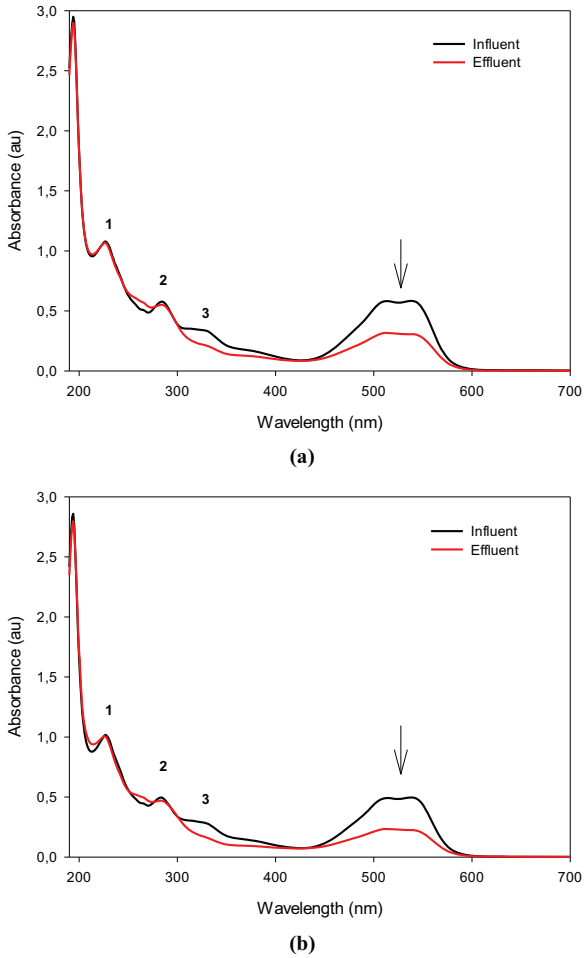


Figure 2. Absorption spectrum during anaerobic (stage II) (a), and microaerobic (stage III) (b) conditions.

3.2. Color Removal Under Anaerobic and Microaearobic Conditions with AQDS Addition

The addition of low concentration of AQDS (25 μM) improved color removal almost immediately, and the average efficiency was 82.5%, around 30% higher compared to stage III ($p < 0.001$). This result is similar to Dos Santos et al.,[23] who used an expanded granular sludge bed (EGSB) reactor (HRT = 10 h) for RR2 removal and obtained an average efficiency of 88%, whereas an efficiency of 56% was obtained when AQDS was absent.

During stage V, without microaeration, the performance was similar to that of the last stage ($p = 0.214$), with an efficiency of 82% (Table 1, **Figure 3**). Therefore, the microaeration did not affect the color removal either in the absence (stages II and III) or in the presence (stages IV and V) of the redox mediator.

Color removal of azo dyes under anaerobic conditions in the presence of redox mediator occurs in two steps: the first one consists in an enzymatic reduction of redox mediator by electrons generated in oxidative processes; and the second one consists in a chemical transfer of electrons to azo dyes, then redox mediator is regenerated.[23] Therefore, the results indicate that the low concentration of oxygen present did not have any

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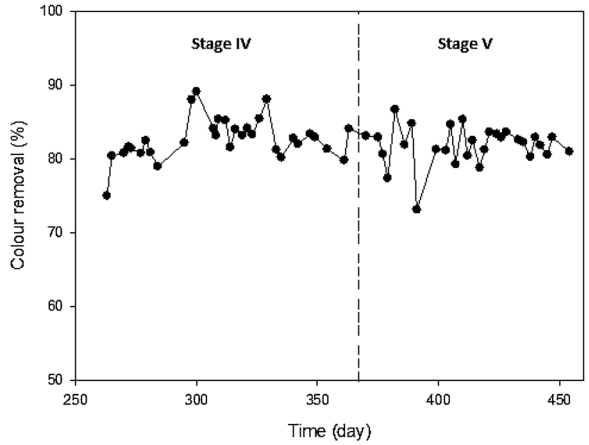


Figure 3. Color removal efficiency during stages VI (anaerobic conditions with AQDS) and V (microaerobic conditions with AQDS).

negative effect on both the low ORP necessary for dye reduction and redox mediator regeneration. The latter finding is important if the application of both microaeration to remove recalcitrant compounds present in textile wastewater and AQDS to speed up color removal is considered. Therefore, the microaeration would not affect negatively the AQDS activity.

The absorption spectrum at stages IV and V showed a higher decrease in absorbance peak (arrow) at 513 nm (**Figure 4**) compared to stages II and III (Figure 2), indicating that a higher dye reduction was taking place.

In relation to the three peaks present in the UV region (Figure 4), no significant differences between stages IV and V were found. However, by comparing with stages II and III, a higher absorbance at peaks 1 and 2 was achieved (Figure 2 and 4), indicating a higher azo dye by-products formation. Moreover, a higher difference between influent and effluent absorbance was observed by analyzing peak 3 (naphthalene ring) (Figure 2 and 4).

3.3. Color Removal Under Anaerobic and Microaearobic Conditions with AQDS Addition in a Reactor Operated in Batch

As the microaeration effect on amines formation and their subsequent mineralization during the continuous-flow experiment was not clear, a batch test was carried out to analyze whether the degradation occurred in a different way inside the reactor. Considering that the HRT of 24 h could be too high, a hypothesis of formation, and mineralization of amines under anaerobic and microaerobic conditions was investigated.

Effluent samples were collected every hour, and the total batch time was 10 h. For anaerobic conditions, results indicate that, at 3 h, there is a peak of a compound with benzene ring (230 nm). After this time, this peak started to decrease until be equal to effluent (after 10 h) (**Figure 5**).

This behavior shows that probably, inside the reactor, formation and mineralization of compounds, due to dye degradation, were occurring. The efficiency was 73.5% after 10 h, close to the average efficiency at stage IV (82.5%). This

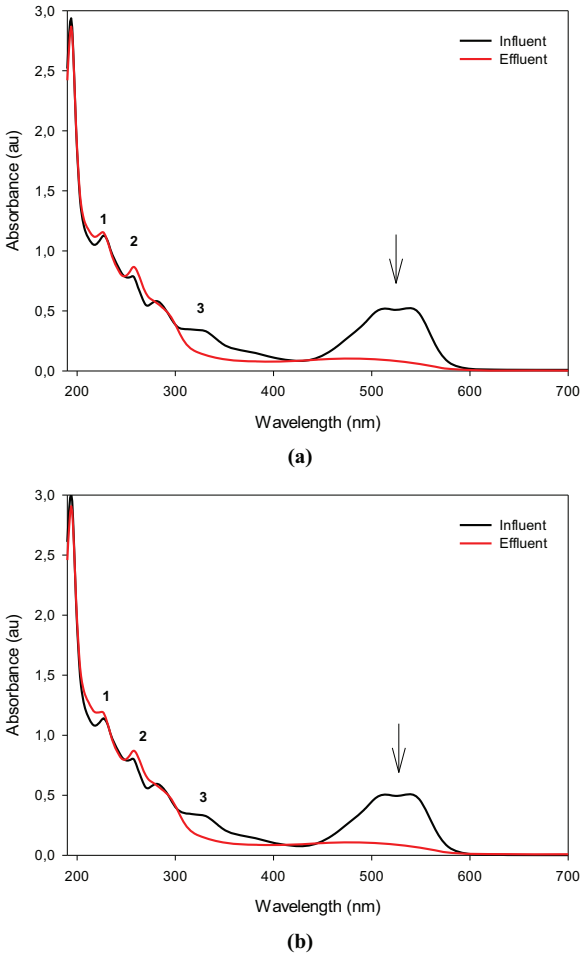


Figure 4. Absorption spectrum during stages IV (anaerobic conditions with AQDS) (a), and V (microaerobic conditions with AQDS) (b).

indicates that it is possible to work with a lower HRT. Under anaerobic conditions, after 10 h, it was possible to reach a color removal close to the one obtained in 24 h (**Figure 6**).

However, under microaerobic conditions, with the same time, the efficiency was 66.7%, approximately 15% less than that obtained in 24 h (82%) (Figure 6). This behavior shows that, probably, due to the establishment of appropriate microaerobic conditions to remove dyes, this system took more time to reach the ORP necessary to remove color. In addition, the formation and reduction of the peaks at both conditions were possible.

3.4. COD Removal and Operational Stability

Operational data for reactors performance are shown in Table 1. The average effluent pH ranged from 7.2 to 7.9, which is close to the recommended values.[16] Pearce et al.[24] reported that the optimum pH for color removal is generally neutral or slightly alkaline.

Average concentration of influent and effluent COD of anaerobic and microaerobic conditions, with and without AQDS, as well as its performance in COD removal is shown in Table 1.

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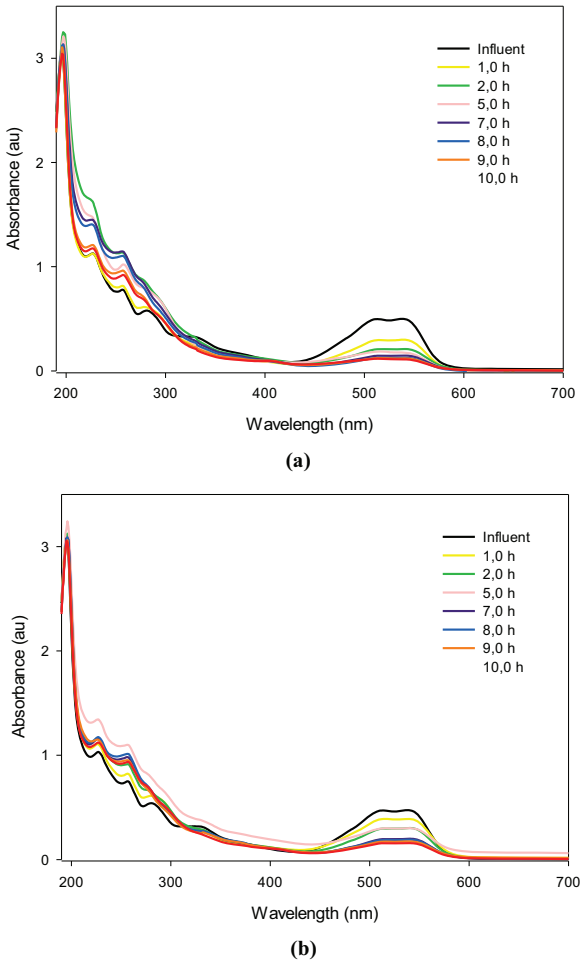


Figure 5. Absorption spectrum of batch experiment under anaerobic (a) and microaerobic (b) conditions.

The difference between the stages related to COD removal is statistically different ($p < 0.001$) for all the stages.

When comparing stages I and II, a slight decrease in COD removal from 93% to 89% can be observed. The addition of oxygen by microaeration caused another decrease in COD removal, from 89% to 86%. The usual slightly positive or negative effect of microaeration on COD removal will depend on many factors such as air flow rate, substrate complexity, microbial community, and others.[25,26]

At stage IV, the redox mediator AQDS was added to the reactor. At this stage, the efficiency increased from 86% to 90%. A different result was obtained by Firmino et al.,[13] who found almost identical efficiency in a reactor containing AQDS (100 μM). Cervantes et al.[27] related that the addition of AQDS (30 μM) increased the average COD removal from 80% to 87%, when using azo acid Orange 7 in a UASB reactor (HRT = 6 h). Albuquerque et al.[28] aimed to decolorize azo Acid Orange 7 (25 mg L^{-1}) in the presence of starch (1150 mg L^{-1}) as co-substrate. The authors observed a decrease in efficiency when a redox mediator (50 μM) was added to an anaerobic–aerobic reactor.

At stage V, the microaeration was turned off, but the AQDS was maintained in the reactor. At this stage, the COD removal

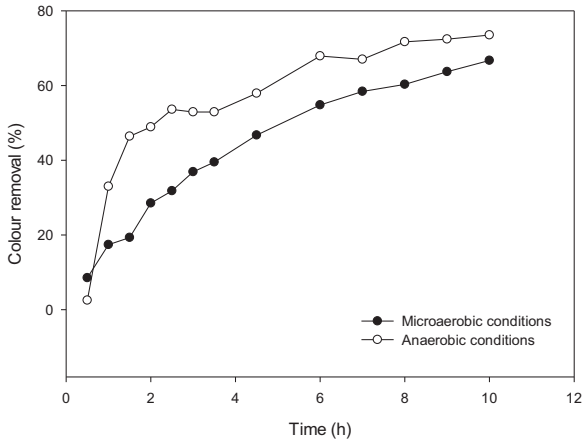


Figure 6. Color removal efficiency during anaerobic and microaerobic batch experiments.

increased slightly from 90% to 93%, similar to the values found between stages IV and V, which indicates that the microaeration promoted a slightly negative impact on organic matter removal.

3.5. Microbial Community

The effect of operational conditions on diversity, evenness, and richness of the bacterial and archaeal communities was assessed by unweighted pair group method with arithmetic mean (UPGMA) clustering analysis based on the Pearson similarity coefficient.[12] Ecological parameters were obtained according to Marzorati et al.:[29] range-weighted richness (R_r), Shannon diversity (H), and functional organization (F_o), for both bacterial and archaeal communities.

The sludge presented the following parameters: for bacteria, $R_r = 69$, $H = 2.65$, and $F_o = 33$; whereas, for archaea, $R_r = 76$, $H = 2.56$, and $F_o = 40$ (**Table 2**), which indicates that the archaeal community was dominant in the inoculum.

The results for bacteria show that, after the start-up stage, R_r was reduced by 20%, H was reduced by ≈6%, while F_o was increased by 27%. These parameters indicate that the bacterial community became slightly more specialized (despite retaining a medium level specialization) than at the previous stage, probably because the use of ethanol as substrate. Archaeal community presented different changes: R_r was increased by 54%, H was increased by ≈15%, whereas F_o showed no significant change (<5%). These changes indicate that generalist archaea were favored, increasing the populational parameters, but maintaining the specialization level at medium range.

At the second stage, the use of azo dye had a positive impact in both domains: for bacteria, R_r was increased by 154%, and H was increased by 21%, whereas, for archaea, Rr was increased by 8%, and H was increased by 6%. F_o for both domains was reduced by 7%. These results indicate that the introduction of the dye favored significantly the bacterial community and slightly the archaea. The domains maintained medium levels of F_o, despite the slight decrease, which indicates that both were colonized by generalists, thus, no dominant specialized groups were present.

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Table 2. Shannon diversity index (H), ranged weighted richness (R_r), and functional organization (F_o) calculated from the DGGE patterns.

		Stage				
Parameter	Seed sludge	I	II	III	IV	V
Bacteria						
F_o (%)	33	42	39	39	35	6
R_r	69	55	140	123	30	101
H	2.65	2.51	3.05	2.93	2.75	2.83
Archaea						
F_o (%)	40	41	38	35	30	36
R_r	76	117	127	74	43	43
H	2.56	2.94	3.09	2.83	2.82	2.84

R_r, low (<10), medium (10–30), high (>30);[29] F_o, low F_o/high evenness (25%); intermediate F_o and evenness (30–70%), high F_o/low evenness (70%);[29] H, 1.5–3.5 (low and high species evenness and richness, respectively).[30]

This result shows that fermentative bacteria are more important to azo dye reduction than archaea, probably because archaea cannot directly reduce the dye, as well as due to competition between methanogenesis and azo dye reduction for equivalents available.[31]

At stage III, microaeration started, promoting different level of impacts for each domain: for bacteria, R_r was decreased by 12%, H and F_o showed no significant change (<5%); for archaea, Rr was decreased by 42%, H was decreased by 8%, and F_o was decreased by 8%. These results indicate that the number of bacterial and archaeal species decreased significantly, despite their distribution presented little to no changes.

Even though there was a quantitative decrease of bacterial and archaea species, there was no significant change in dye removal because the required quantity of electrons for color removal was too low.[32]

At stage IV, the addition of AQDS under microaerobic conditions promoted further impacts than those observed at the previous stage: for bacteria, R_r was decreased by 75%, H was decreased by 6%, and F_o was decreased by 10%; for archaea, R_r was decreased by 42%, H showed no significant change (<5%), and F_o was decreased by 14%. The significant impacts observed for both domains on R_r, despite the slight changes in H and F_o, once again indicate that the carrying capacity was reduced, thus reducing the number of species, but not their distribution.

Considering that, after the microaeration, R_r was decreased at both stages (III and IV), it can be inferred that the anaerobic species were probably the dominant over the facultative species, since no populational recovery was observed after the introduction of oxygen.

Although there was a huge impact on the microbial community, the dye removal efficiency increased, indicating that the capacity of electron transfer of the system was improved by the addition of the redox mediator AQDS.

At stage V, the reactor returned to anaerobic condition, thus promoting a new impact in both domains: the results for bacteria showed that the anaerobic condition promoted increase in populational parameters, as R_r was increased by 236%, H showed no significant change (<5%), whereas F_o was decreased by 83%. The results for archaea revealed a completely different outcome, as R_r and H showed no significant change (<5%), whereas F_o was increased by 20%. The number of species increased drastically, which indicates that the carrying capacity for bacteria increased. Furthermore, maintaining H and decreasing F_o indicate that the ecological niches were maintained, and the distribution of species increased, despite the majority being generalists, thus not specialized in dye degradation. This confirms that reduction of the azo bond is not an inherent property of methanogens.[32]

4. Conclusions

Considering the conditions applied to the present experiment, it was not possible to show any microaeration effect on dye reduction and by-products formed.

Low concentration of oxygen did not have any negative effect on both dye reduction and redox mediator regeneration. Because of that, AQDS considerably enhanced RR2 reduction.

It seems that the cleavage of the double bond between nitrogen groups is a much easier way to start the attack of the azo dye, and it is independent of the presence of oxygen. Thus, the introduction of hydroxyl groups into the aromatic ring was not playing a role in the process of color removal. However, microaeration would be important to textile wastewater treatment, in which other recalcitrant compounds under anaerobic conditions would be present.

In terms of ecological parameters, the ecological niches were maintained, and the distribution of species increased, despite the majority being generalists, thus not specialized in dye degradation. Furthermore, anaerobic species are dominant over facultative species, which explains why there was no population diversity recovery after the microaeration started.

Abbreviations

AOX, adsorbable organic halide; AQDS, anthraquinone-2,6-disulfonate; AQS, anthraquinone-2-sulfonate; COD, chemical oxygen demand; EGSB, expanded granular sludge bed; F_o, functional organization; H, Shannon diversity; HRT, hydraulic retention time; ORP, oxidation reduction potential; Q_{inf}/Q_{air} influent flow rate to air flow rate ratio; R_r range-weighted richness; RR2, Reactive Red 2; TA, total alkalinity; UASB, up-flow anaerobic sludge blanket; UPGMA, unweighted pair group method with arithmetic mean; VFA, volatile fatty acid; VSS, volatile suspended solids.

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Conflict of Interest

The authors have declared no conflict of interest.

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Keywords

anaerobic treatment, AQDS, aromatic amines mineralization, microaerobic system

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